

ZHANGIZTOBE, KazakhstanWMO: **363970**Lat: **49.217N**Lon: **81.217E**Elev: **455**StdP: **95.98**Time Zone: **6.00 (E06)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.5	-31.9	-38.4	0.1	-35.5	-34.8	0.1	-31.8	24.4	-7.7	23.8	-8.3	0.9	190	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	33.2	17.6	31.2	17.2	29.5	16.6	19.3	28.4	18.6	27.5	17.9	26.4	3.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.5	21.6	15.7	11.7	21.3	14.7	11.1	21.0	57.1	28.5	54.5	27.7	52.4	26.3	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	17.4	15.9	DB	-35.6	35.9	4.7	1.7	-38.9	37.1	-41.7	38.1	-44.3	39.1	-47.7	40.3	(4)
(5)			WB	-35.2	21.3	4.6	1.2	-38.5	22.2	-41.3	22.9	-43.9	23.6	-47.2	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-15.8	-13.6	-5.2	7.5	13.9	20.2	21.4	19.7	13.3	5.7	-4.2	-12.0		
	DBStd	14.61	7.92	7.66	7.26	5.72	4.76	3.85	3.36	3.80	5.35	5.39	7.87	7.52		
	HDD10.0	3349	800	661	471	109	18	0	0	0	27	154	425	683		
	HDD18.3	5397	1058	894	729	325	148	27	11	29	166	393	675	942		
	CDD10.0	1281	0	0	0	34	141	305	354	301	126	20	0	0		
	CDD18.3	288	0	0	0	0	12	82	107	72	16	0	0	0		
	CDH23.3	3315	0	0	0	12	209	867	1096	882	246	3	0	0		
	CDH26.7	1203	0	0	0	1	56	304	420	343	80	0	0	0		
(14) Wind	WSAvg	4.4	7.1	6.6	5.1	3.4	3.4	2.9	2.8	2.6	3.0	3.8	5.4	7.0		
(15) Precipitation	PrecAvg	314	16	15	19	24	29	41	52	24	18	25	30	22		
	PrecMax	442	27	31	45	75	68	107	149	87	46	63	52	49		
	PrecMin	194	3	5	1	4	6	13	2	2	1	6	3	6		
	PrecStd	55	7	7	12	14	15	23	30	18	12	14	11	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.5	1.8	12.7	25.2	31.6	34.1	35.9	35.7	32.0	23.2	13.3	2.1		
		MCWB	-2.2	0.0	6.7	13.9	17.0	17.1	19.4	17.6	16.4	11.3	6.7	-0.3		
	2%	DB	-3.1	-0.4	9.2	22.1	27.9	31.7	32.9	32.8	29.1	19.7	9.7	0.5		
		MCWB	-4.5	-2.0	5.0	12.2	15.9	17.4	18.1	17.5	14.7	9.9	5.0	-1.1		
	5%	DB	-4.6	-2.2	6.3	19.6	25.5	30.0	30.8	30.3	26.1	16.7	7.1	-1.6		
		MCWB	-6.0	-3.7	3.3	11.1	14.4	17.0	17.5	16.5	13.6	8.5	3.7	-2.9		
	10%	DB	-6.2	-4.0	3.9	17.1	23.3	28.2	29.0	28.2	23.3	13.9	4.6	-3.3		
		MCWB	-7.4	-5.4	1.6	9.6	13.4	16.6	17.0	15.8	12.4	7.5	1.9	-4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.0	0.4	7.7	15.1	18.8	19.9	20.7	19.8	16.9	12.3	7.9	0.3		
		MCDB	-0.6	1.5	11.9	24.1	28.5	28.4	29.0	30.3	30.5	19.9	11.8	1.8		
	2%	WB	-4.4	-1.8	5.2	13.1	16.7	19.0	19.5	18.4	15.4	10.5	5.6	-1.1		
		MCDB	-3.3	-0.5	8.6	20.6	25.4	28.0	28.2	28.9	27.0	17.8	8.8	0.3		
	5%	WB	-5.9	-3.6	3.3	11.7	15.3	18.3	18.7	17.6	14.2	9.2	3.7	-2.9		
		MCDB	-4.7	-2.4	6.2	18.4	23.4	26.8	27.3	27.2	24.3	15.7	7.0	-1.6		
	10%	WB	-7.5	-5.4	1.6	10.3	14.2	17.5	18.1	16.8	13.0	7.9	2.0	-4.5		
		MCDB	-6.2	-4.1	3.7	16.1	21.7	25.6	26.1	25.6	22.0	13.5	4.6	-3.4		
(35) Mean Daily Temperature Range	MDBR		7.0	8.1	8.3	11.6	13.0	12.1	11.6	13.2	13.4	10.3	7.2	6.6		
	5% DB	MCDBR	6.8	7.6	10.5	15.6	16.2	14.7	14.8	16.3	17.3	14.7	9.6	6.9		
		MCWBR	6.2	6.8	7.2	8.2	7.6	5.0	4.8	5.7	7.6	6.0	6.0	6.2		
	5% WB	MCDBR	7.2	7.5	10.0	13.6	14.2	12.3	12.1	13.8	15.5	13.1	9.2	6.9		
		MCWBR	6.6	7.0	7.2	7.8	7.3	5.0	4.8	5.6	7.2	7.4	6.4	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.291	0.326	0.380	0.415	0.407	0.408	0.426	0.392	0.350	0.351	0.331	0.298		
	taud		2.075	2.008	2.015	2.104	2.188	2.261	2.223	2.294	2.357	2.253	2.142	2.049		
	Ebn at Noon		727	789	810	827	853	853	830	843	846	764	672	650		
	Edh at Noon		87	119	142	145	139	130	134	119	100	93	82	78		
(44) All-Sky Solar Radiation	RadAvg		1.67	2.79	4.09	4.96	6.38	6.71	6.20	5.74	4.32	2.49	1.39	1.21		
	RadStd		0.08	0.13	0.32	0.39	0.38	0.41	0.41	0.30	0.35	0.23	0.12	0.11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	-0.44	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page